

TEMPORALLY CENTERED SIGREG IMPROVES MULTI-TASK LEWORLDMODEL LEARNING: FROM ANALYSIS TO METHOD

Chang Liu Fei Suo Yanzhou Jin
Yusuke Iwasawa Yutaka Matsuo Yaonan Zhu[†]

Graduate School of Engineering, The University of Tokyo, Tokyo, Japan

[†]Corresponding author: yaonan.zhu@weblab.t.u-tokyo.ac.jp

ABSTRACT

Recent work on LeWorldModel (LeWM) has shown that the Sketched Isotropic Gaussian Regularizer (SIGReg) enables stable end-to-end world-model learning from pixels by regularizing the latent marginal distribution toward an isotropic Gaussian, thereby preventing representation collapse. While effective and elegant in single-task settings, this recipe does not extend reliably to multi-task training, leading to substantially worse downstream behavior-cloning performance. In this paper, we show that marginal Gaussianization compresses the separation between task-dependent latent clusters relative to within-cluster variation. This compression introduces representation **aliasing** across tasks and states, and makes the learned representations highly sensitive to small visual perturbations. To address this problem, we apply SIGReg to **temporally centered residuals** rather than to the latent marginal distribution. This surrogate target places no direct regularization pressure on the separation among cluster centers, removes the requirement that the full latent follow a single isotropic Gaussian, and retains the anti-collapse effect of SIGReg. On the LIBERO benchmark, our method improves downstream success on the long-horizon suite by **1.7 $\times$** and raises the average success rate across four suites from 53.2% to **73.6%**. Without external pretraining, it slightly outperforms Diffusion Policy trained from scratch and approaches the performance of large-scale pretrained policy baselines. These results reveal a structural incompatibility between marginal Gaussian priors and multi-task latent structure, and provide a simple route toward stable and **scalable** end-to-end multi-task world-model learning. [Project Page](#).

1 INTRODUCTION

World models (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020) learn predictive latent representations of environment dynamics, allowing agents to anticipate the consequences of their actions and enabling planning and policy learning. Many existing approaches (Hafner et al., 2019; 2020; 2025) rely on observation reconstruction, reward prediction, or other task-specific supervision. Although effective, these objectives may retain unnecessary visual details and complicate learning from large reward-free multi-task datasets (Deng et al., 2022; Balestriero & LeCun, 2024; Hutson et al., 2024). Reconstruction-free joint-embedding predictive learning (Assran et al., 2023; 2025) offers an appealing alternative. LeWorldModel (LeWM) (Maes et al., 2026) jointly trains an observation encoder and an action-conditioned latent predictor directly from pixels. To prevent representation collapse, it uses the Sketched Isotropic Gaussian Regularizer (SIGReg) (Balestriero & LeCun, 2025), which applies an Epps–Pulley (EP) normality criterion (Epps & Pulley, 1983) to random latent projections and encourages the latent marginal to follow an isotropic Gaussian. This enables end-to-end world-model learning without reconstruction, pretrained encoders, momentum targets, or reward supervision.

However, global non-collapse is not sufficient for multi-task control. On the LIBERO-Long suite (Liu et al., 2023a), the average downstream success rate of Raw LeWM falls from 40.0% under in-

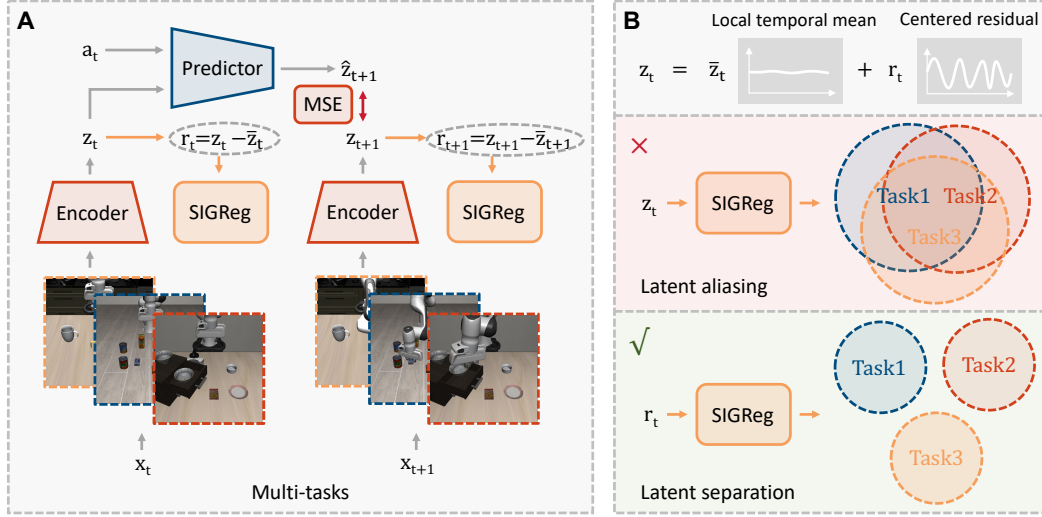


Figure 1: Overview of TC-LeWM. (A) The encoder and action-conditioned predictor learn latent dynamics, while TC-LeWM decomposes each latent as $z_t = \bar{z}_t + r_t$ and applies SIGReg to the centered residual r_t . (B) Marginal SIGReg contracts task-dependent structure, whereas TC-SIGReg removes this direct pressure while retaining anti-collapse regularization.

dependent single-task training to 29.9% under 10-task joint training, with eight of ten tasks degrading. Although the representation remains statistically non-collapsed, its geometry becomes poorly aligned with the state distinctions required by a downstream behavior-cloning policy.

We trace this degradation to a conflict between marginal Gaussianization and multi-task latent structure. A controlled Monte Carlo analysis of randomized multi-component mixtures shows that the EP objective exerts contractive pressure on component-center separation relative to within-component variation. This can induce representation aliasing across tasks and states despite global non-collapse.

To address this issue, we introduce *Temporally Centered LeWorldModel* (TC-LeWM), illustrated in Fig. 1. TC-LeWM applies SIGReg to temporally centered residuals rather than to the full latent marginal. This simple change removes direct Gaussianization pressure on latent structure while retaining the anti-collapse effect of SIGReg. Across the four LIBERO suites, TC-LeWM raises the suite-wise 10-task average success rate from 53.2% to 73.6%. Under unified 40-task training, it maintains 73.5% average success, compared with 44.4% for Raw LeWM. The learned representations also exhibit multimodal, task- and progress-dependent organization and greater robustness.

Contributions. (i) We identify a multi-task failure mode of marginal SIGReg and show that the EP gradient contracts component-center separation, leading to representation aliasing despite global non-collapse. (ii) We propose TC-SIGReg, which regularizes temporally centered residuals to remove direct Gaussianization pressure on latent structure while retaining anti-collapse. (iii) Experimentally, TC-LeWM consistently improves downstream control from single-task to unified 40-task LIBERO training and learns more organized and perturbation-robust representations.

2 RELATED WORK

Latent world models. World models learn compact predictive representations of environment dynamics for planning and policy learning (Ha & Schmidhuber, 2018; Hafner et al., 2020; Schrittwieser et al., 2020; Hafner et al., 2021; 2025). Early latent-dynamics approaches showed that environment dynamics could be learned and used for control in compact latent spaces (Watter et al., 2015; Hafner et al., 2019). Subsequent stochastic state-space models combined deterministic memory with latent uncertainty, enabling control directly from pixel observations (Hafner et al., 2020; 2019). Dreamer-style methods (Hafner et al., 2020; Wu et al., 2023; Hafner et al., 2025) further

learned policies from trajectories imagined within the world model and extended this paradigm across continuous control, Atari, diverse domains, and physical robots.

Most of these methods rely on observation reconstruction, reward prediction, or other task-specific objectives. Although effective, reconstruction objectives may retain control-irrelevant visual information, while reward-dependent objectives require supervision unavailable in reward-free datasets (Fu et al., 2021; Wan et al., 2023; Zhu et al., 2023; Balestriero & LeCun, 2024; Hutson et al., 2024; Wang et al., 2024; Sobal et al., 2025).

Joint-embedding predictive learning. Joint-Embedding Predictive Architectures (JEPAs) learn representations by predicting target embeddings directly in latent space rather than reconstructing raw observations (Assran et al., 2023; Dawid & LeCun, 2024). I-JEPA (Assran et al., 2023) instantiated this paradigm for images, and V-JEPA (Bardes et al., 2024; Assran et al., 2025) extended it to spatiotemporal representation learning from video. Related approaches have adapted latent prediction to action-conditioned world modeling: DINO-WM (Zhou et al., 2025) learns dynamics over frozen pretrained visual features for zero-shot planning, whereas PLDM (Sobal et al., 2025) jointly learns reconstruction-free latent dynamics from reward-free trajectories. However, non-contrastive joint-embedding objectives admit collapsed solutions, motivating mechanisms such as momentum target encoders, stop-gradient asymmetry, explicit variance regularization, and redundancy-reduction objectives (Grill et al., 2020; Chen & He, 2021; Bardes et al., 2022; Zbontar et al., 2021).

LeWorldModel and multi-task learning. LeWM (Maes et al., 2026) enables end-to-end reconstruction-free world-model learning directly from pixels with only a single hyperparameter to tune. It jointly trains an encoder and an action-conditioned predictor using latent next-state prediction together with SIGReg (Balestriero & LeCun, 2025), which prevents representation collapse without pretrained encoders, momentum targets, observation reconstruction, or reward supervision.

Existing evaluations primarily consider separately trained models or individual task settings, leaving the behavior of marginal SIGReg under joint multi-task training unclear. We show that a LeWM representation can remain globally non-collapsed while compressing or entangling task- and state-dependent structure needed for downstream control. Our work analyzes this failure mode and replaces marginal regularization with a temporally centered residual target.

3 WHY MARGINAL SIGREG CAN DEGRADE MULTI-TASK LEWM LEARNING

In this section, we first document the degradation of raw LeWM under multi-task training, then use a controlled mixture analysis to characterize the contractive pressure exerted by marginal SIGReg, and finally discuss how the resulting latent overlap can hinder downstream behavior-cloning policies.

3.1 DEGRADATION OF LEWM UNDER MULTI-TASK TRAINING

Multi-task training and downstream evaluation. Consider a task set $\mathcal{T} = \{1, \dots, N\}$, where task i is associated with a demonstration dataset $\mathcal{D}_i$. Raw LeWM jointly trains an encoder and an action-conditioned latent predictor on the aggregated dataset $\mathcal{D} = \bigcup_{i=1}^N \mathcal{D}_i$ using

$$\mathcal{L}_{\text{Raw}} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{SIGReg}}(\mathcal{Z}), \quad \mathcal{Z} = \{\mathbf{z}_t \mid \mathbf{z}_t = f_{\theta}(\mathbf{x}_t)\}. \quad (1)$$

After world-model training, the encoder is frozen and evaluated using a task-conditioned flow-matching behavior-cloning policy $\pi_{\phi}(\mathbf{a}_t \mid \mathbf{z}_t, i)$.

Multi-task degradation. In Section 5.3, we compare independent single-task training with joint training on the same ten LIBERO-Long tasks. Raw LeWM achieves an average success rate of 40.0% when a separate pipeline is trained for each task, but only 29.9% when all ten tasks share one encoder, predictor, and task-conditioned policy. This corresponds to a decrease of 10.1 percentage points, with eight of the ten tasks deteriorating under joint training. The result shows that global non-collapse alone does not ensure a representation suitable for shared multi-task control. In the following sections, we analyze the source of this degradation.

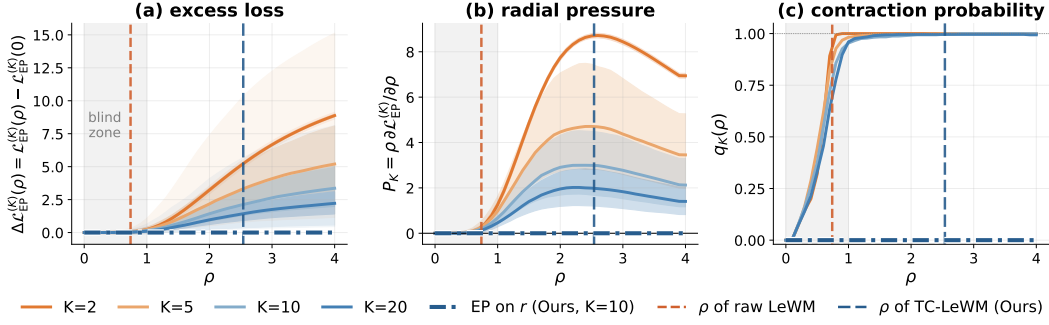


Figure 2: Monte Carlo analysis of the EP objective on randomized multi-component mixtures. (a) Excess EP loss relative to $\rho = 0$. (b) Median radial pressure over randomized configurations; positive values indicate contractive pressure on the global component-center spread. (c) Fraction of randomized configurations with detectable contractive pressure. Vertical dashed lines mark the measured ρ values of Raw LeWM and TC-LeWM on LIBERO-Long.

3.2 MARGINAL SIGREG EXERTS CONTRACTIVE PRESSURE ON MULTI-TASK STRUCTURE

Latent decomposition and mixture model. To analyze the source of the multi-task degradation, we decompose each latent representation into a local temporal mean and a complementary short-timescale residual:

$$\bar{\mathbf{z}}_t \triangleq \frac{1}{|\mathcal{W}_t|} \sum_{s \in \mathcal{W}_t} \mathbf{z}_s, \quad \mathbf{r}_t \triangleq \mathbf{z}_t - \bar{\mathbf{z}}_t, \quad \mathbf{z}_t \equiv \bar{\mathbf{z}}_t + \mathbf{r}_t, \quad (2)$$

where $\mathcal{W}_t$ denotes a local temporal window. The local mean captures low-frequency task- and context-dependent structure, while the residual captures complementary short-timescale variation.

SIGReg applies the Epps–Pulley (EP) criterion to random one-dimensional projections of the latent marginal. For a unit projection vector $\mathbf{w}$, linearity preserves the temporal decomposition:

$$x = \mathbf{w}^\top \mathbf{z}_t = \underbrace{\mathbf{w}^\top \bar{\mathbf{z}}_t}_c + \underbrace{\mathbf{w}^\top \mathbf{r}_t}_\epsilon. \quad (3)$$

We therefore study the EP objective through the projected variables c and ϵ , modeling c as a balanced K -component Gaussian mixture and $\epsilon \sim \mathcal{N}(0, \sigma_r^2)$ as independent Gaussian variation. This yields

$$p(x) = \frac{1}{K} \sum_{k=1}^K \mathcal{N}(x; \mu_k, \sigma^2), \quad \sigma^2 = \tau^2 + \sigma_r^2, \quad (4)$$

where μ_k is the projected center of the k -th component and σ is the common within-component standard deviation. The balanced, homoscedastic construction isolates the effect of component-center geometry on the EP objective.

Randomized mixture diagnostic. For each K , we generate $M = 500$ randomized component-center configurations and vary their normalized global spread $\rho = \text{std}_k[\mu_k]/\sigma$, where σ is the common within-component standard deviation. We evaluate $K \in \{2, 5, 10, 20\}$ and $\rho \in \{0, 0.1, \dots, 4.0\}$ using a shape-only EP diagnostic that controls for global location and scale. Full sampling and estimation details are provided in Appendix A.2.

Results and implications for policy learning. Figure 2 summarizes the Monte Carlo results. Panel (a) shows that the excess EP loss increases with the normalized component-center spread ρ , indicating that EP increasingly penalizes separated multi-component structure. Panel (b) examines the corresponding radial gradient signal, which becomes consistently positive outside the small- ρ weak-gradient region. A positive signal induces a component of the gradient-descent update that reduces ρ and contracts the global center spread. Panel (c) further shows that the fraction of randomized configurations with detectable contractive pressure increases with ρ , indicating that this tendency is systematic across different center geometries. Together, these results show that marginal EP regularization opposes well-separated multi-component structure.

As the mixture approaches $\rho \lesssim 1$, the global center spread becomes comparable to the within-component standard deviation, making component overlap and representation aliasing likely, as illustrated in Fig. 1(B). This experimental observation is consistent with the latent geometries examined in benchmark LIBERO-Long. Raw LeWM yields $\rho = 0.74$, placing its learned representation in this low-separation region, whereas TC-LeWM yields $\rho = 2.54$. This contrast is consistent with the latent geometries examined in Section 5.4.

For a task-conditioned behavior-cloning policy $\pi_\phi(\mathbf{a}_t | \mathbf{z}_t, i)$, the task identity can resolve ambiguity attributable solely to the task label, but cannot recover within-task state distinctions already aliased in $\mathbf{z}_t$. Similar latent codes may therefore correspond to incompatible actions, providing a mechanism consistent with the multi-task degradation observed in Table 2.

4 HOW TO FIX THE DEGRADATION: FROM MARGINAL TO RESIDUAL

We introduce Temporally Centered LeWorldModel (TC-LeWM), which applies SIGReg to temporally centered residuals rather than to the full latent marginal.

4.1 TEMPORALLY CENTERED LEWORLDMODEL

Model and training objective. As illustrated in Fig. 1, TC-LeWM retains the LeWM encoder and action-conditioned latent predictor, $\mathbf{z}_t = f_\theta(\mathbf{x}_t)$ and $\hat{\mathbf{z}}_{t+1} = g_\psi(\mathbf{z}_t, \mathbf{a}_t)$. For each latent sequence, it computes

$$\bar{\mathbf{z}}_t = \frac{1}{|\mathcal{W}_t|} \sum_{s \in \mathcal{W}_t} \mathbf{z}_s, \quad \mathbf{r}_t = \mathbf{z}_t - \bar{\mathbf{z}}_t, \quad (5)$$

where $\mathcal{W}_t$ is a local temporal window. Let $\mathcal{R} = \{\mathbf{r}_t\}$ denote the residuals in a training batch. TC-LeWM optimizes

$$\mathcal{L}_{\text{TC}} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{SIGReg}}(\mathcal{R}). \quad (6)$$

Compared with Raw LeWM, the change lies in the regularization target: from the full latent marginal $\mathcal{Z}$ to the temporally centered residual $\mathcal{R}$.

After world-model training, the encoder is frozen and used to train a task-conditioned flow-matching behavior-cloning policy $\pi_\phi(\mathbf{a}_t | \mathbf{z}_t, i)$. Raw LeWM and TC-LeWM use identical downstream policy protocols. Architecture details are provided in Appendix A.1.

4.2 WHY THE RESIDUAL TARGET WORKS

TC-SIGReg has two complementary properties: it removes the direct contractive pressure on low-frequency multi-task structure while retaining an effective anti-collapse constraint.

No direct pressure on low-frequency separation. Under the decomposition in Section 3.2, a projected latent variable satisfies $x = c + \epsilon$, where c represents the local temporal mean and ϵ the centered residual. Changing ρ scales the component centers of c while leaving the distribution of ϵ unchanged. The residual EP objective is therefore invariant to ρ under this construction and places no direct regularization pressure on the separation of the low-frequency component centers. In Fig. 2, the residual-target results are shown by the blue dash-dotted curves. Both the excess loss and radial gradient remain close to zero as ρ varies. TC-SIGReg thus avoids the center-contraction mechanism of marginal SIGReg without explicitly forcing the task-dependent clusters apart.

Anti-collapse property. Regularizing the residual still penalizes temporally collapsed representations. If $\mathbf{z}_t$ becomes constant within a temporal window, then $\mathbf{r}_t \equiv \mathbf{0}$, and every one-dimensional projection of the residual follows the point mass δ_0 , in which case EP will penalize the degenerate distribution. The prediction and residual-regularization objectives provide complementary constraints. Temporal collapse incurs a large EP penalty, whereas unstructured residual noise may satisfy the marginal regularizer but is generally difficult to predict from the current latent and action. Their combination therefore favors non-degenerate, predictable temporal variation, retaining effective anti-collapse pressure without directly Gaussianizing low-frequency multi-task structure.

Table 1: Closed-loop success rates (%) under suite-wise 10-task training. Diffusion Policy, Octo, and OpenVLA are published mean results reported in Kim et al. (2025) and are included as contextual references. Raw LeWM and TC-LeWM are reported as mean $\pm$ std over three full-pipeline training seeds.

Method	Spatial	Object	Goal	Long	Mean
Diffusion Policy (scratch)	78.3	92.5	68.3	50.5	72.4
Octo (fine-tuned)	78.9	85.7	84.6	51.1	75.1
OpenVLA (fine-tuned)	84.7	88.4	79.2	53.7	76.5
Raw LeWM	61.6 $\pm$ 4.3	46.4 $\pm$ 12.8	74.8 $\pm$ 6.2	29.9 $\pm$ 8.3	53.2
TC-LeWM (Ours)	68.6$\pm$1.8	87.8$\pm$3.2	86.0$\pm$1.1	51.8$\pm$9.2	73.6

5 EXPERIMENTS

We evaluate TC-LeWM on the LIBERO benchmark and analyze the performance, geometry, and perturbation robustness of the learned representations.

5.1 EXPERIMENTAL SETUP

Benchmark. LIBERO contains four suites—Spatial, Object, Goal, and Long—each with ten manipulation tasks. Each observation contains two RGB camera views. A shared ViT encoder (Dosovitskiy et al., 2021) processes the two views independently.

World-model pretraining and downstream policy learning. We evaluate the learned representations using a two-stage pipeline. First, the encoder and action-conditioned latent predictor are trained using the world-model objective. During this stage, SIGReg is applied only to the CLS-token representations of the two camera views; the patch tokens are not directly regularized.

After world-model training, the encoder is frozen and a task-conditioned flow-matching behavior-cloning policy is trained on the resulting visual representations: $\pi_\phi(\mathbf{a}_t \mid \mathbf{z}_t, i)$. For downstream policy learning, we use both the CLS token and the patch tokens from each camera view. We empirically find that augmenting the CLS tokens with patch tokens provides more informative spatial features for manipulation than using the CLS tokens alone.

Although the original LeWM evaluates its representations through latent-space planning, behavior cloning serves here as a controlled downstream evaluation protocol. Raw LeWM and TC-LeWM use identical frozen-encoder and downstream-policy settings.

Task aggregation settings. We consider three levels of task aggregation. In the **single-task** setting, one world model and one downstream policy are trained independently for each task in LIBERO-Long. In the **10-task** setting, the ten tasks within each suite share one encoder, latent predictor, and task-conditioned policy. In the **40-task** setting, all four suites share a single pipeline.

Compared methods. We mainly compare **Raw LeWM**, which applies SIGReg to the marginal CLS-token representation $\mathbf{z}_t$, with **TC-LeWM**, which applies the same regularization criterion to the temporally centered residual $\mathbf{r}_t = \mathbf{z}_t - \bar{\mathbf{z}}_t$. The encoder and predictor architectures, prediction objective, training data, optimization settings, and downstream policy protocol are identical across the two methods. TC-LeWM uses a temporal window of $W = 8$ frames, corresponding to approximately 1.4 s on LIBERO. An ablation study on window size is provided in Appendix A.5.

Evaluation. We report closed-loop task success rates. Results are reported as mean $\pm$ std over three training seeds, each evaluated with three seeds and 50 rollouts per task per evaluation seed.

5.2 MAIN RESULTS ON THE LIBERO BENCHMARK

We first evaluate suite-wise 10-task joint training, where all tasks within each LIBERO suite share one world model and downstream policy. Table 1 shows that TC-LeWM improves success over Raw LeWM across all four suites, increasing the average from 53.2% to 73.6%. The largest gains occur on LIBERO-Object and LIBERO-Long (+41.4 and +21.9 points), supporting temporally centered residuals as a more suitable SIGReg target for multi-task world-model learning.

Table 2: Per-task closed-loop success rates (%) on LIBERO-Long under independent single-task and suite-wise 10-task joint training. Within each method, Δ denotes Joint–Single; the final row compares TC-LeWM and Raw LeWM under single-task training.

Method	Setting	T0	T1	T2	T3	T4	T5	T6	T7	T8	T9	Avg
Raw LeWM	Single	32.9	34.7	70.0	68.4	28.2	46.2	16.7	28.2	27.6	47.3	40.0
	Joint	17.3	26.0	48.7	47.6	18.4	42.4	17.1	32.0	9.3	40.0	29.9
	Δ	−15.6	−8.7	−21.3	−20.9	−9.8	−3.8	+0.4	+3.8	−18.2	−7.3	−10.1
TC-LeWM (Ours)	Single	33.3	34.0	84.0	70.7	42.4	69.1	24.0	44.2	31.8	62.7	49.6
	Joint	44.9	40.2	81.6	69.1	60.4	63.3	34.7	32.0	23.3	68.2	51.8
	Δ	+11.6	+6.2	−2.4	−1.6	+18.0	−5.8	+10.7	−12.2	−8.4	+5.6	+2.2
TC – Raw (Single)		+0.4	−0.7	+14.0	+2.2	+14.2	+22.9	+7.3	+16.0	+4.2	+15.3	+9.6

Table 3: Closed-loop success rates (%) under unified 40-task training. All four LIBERO suites share one encoder, predictor, and task-conditioned policy. Suite-level results are reported as mean $\pm$ standard deviation over three full-pipeline training seeds, each evaluated using three evaluation seeds with 50 rollouts per task. The final column reports the change relative to suite-wise 10-task training in Table 1.

Method	Spatial	Object	Goal	Long	Mean	vs. 10-task
Raw LeWM	40.8 $\pm$ 12.0	49.2 $\pm$ 7.6	58.6 $\pm$ 11.0	28.8 $\pm$ 2.0	44.4	−8.8
TC-LeWM (Ours)	72.3 $\pm$ 0.9	83.6 $\pm$ 5.1	84.9 $\pm$ 1.3	53.3 $\pm$ 1.5	73.5	−0.1
Δ (TC – Raw)	+31.5	+34.4	+26.3	+24.5	+29.1	

Without external pretraining, TC-LeWM also slightly exceeds the published from-scratch Diffusion Policy (Chi et al., 2023) average (73.6% versus 72.4%) and approaches fine-tuned Octo (Ghosh et al., 2024) and OpenVLA (Kim et al., 2025) (75.1% and 76.5%). These published results are included only as contextual references because their experimental pipelines differ from ours.

5.3 COMPARISON UNDER INCREASING TASK SCALE

We next examine how the advantage of TC-LeWM changes as the number and diversity of jointly trained tasks increase.

From single-task to 10-task joint training. We first compare independent single-task training with joint training on the same ten LIBERO-Long tasks.

As shown in Table 2, Raw LeWM exhibits clear negative transfer under 10-task joint training: its average success rate decreases from 40.0% to 29.9%, with eight of the ten tasks deteriorating. In contrast, TC-LeWM improves from 49.6% to 51.8%, corresponding to an average change of +2.2 percentage points compared with −10.1 points for Raw LeWM. This result shows that TC-LeWM is substantially more robust to the interference introduced by joint training.

TC-LeWM also outperforms Raw LeWM on nine of the ten tasks under single-task training, improving the average success rate by 9.6 percentage points. This result suggests that relaxing the overly restrictive inductive bias of Gaussianizing the full latent marginal benefits representation learning even in the absence of cross-task interference.

Scaling from suite-wise 10-task to unified 40-task training. We next train a single world model and downstream policy jointly on all forty tasks from the four LIBERO suites.

Table 3 shows that the advantage of TC-LeWM widens under unified 40-task training. Relative to suite-wise 10-task training, Raw LeWM decreases from 53.2% to 44.4% (−8.8 points), whereas TC-LeWM remains nearly unchanged, from 73.6% to 73.5% (−0.1 points). The TC–Raw gap increases from 20.4 to 29.1 points, with TC-LeWM outperforming Raw LeWM by 24.5–34.4 points across all four suites. This scaling behavior is consistent with marginal SIGReg becoming increasingly restrictive when a single representation must accommodate more heterogeneous task-dependent structure. This result further supports the scalability of TC-LeWM to larger multi-task settings.

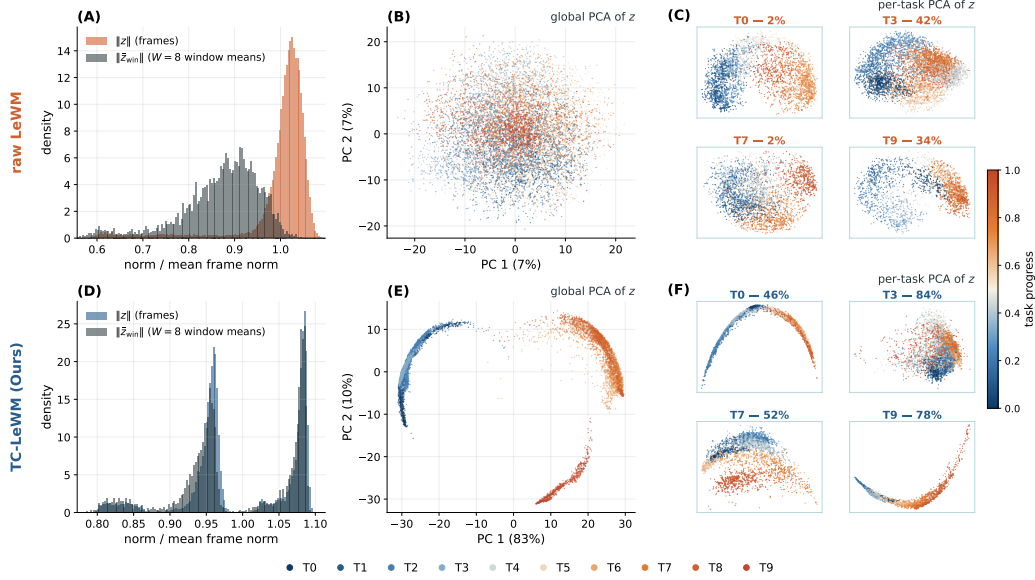


Figure 3: Latent representation analysis on LIBERO-Long. Top: Raw LeWM; bottom: TC-LeWM. (A,D) Normalized distributions of frame-level and window-mean latent norms. (B,E) Global PCA projections colored by task identity. (C,F) Per-task PCA projections for four tasks, colored by normalized task progress; titles report 50-rollout success rates of the visualized training seed. TC-LeWM exhibits reduced short-timescale variation and more organized inter- and intra-task latent structure than Raw LeWM.

5.4 LATENT STRUCTURE AND PERTURBATION ROBUSTNESS

We next examine how the SIGReg target affects latent-space organization and robustness to visual perturbations.

Temporal structure and latent organization. Figure 3 compares Raw LeWM and TC-LeWM using frozen encoders trained jointly on the ten LIBERO-Long tasks. For each observation, we concatenate the CLS features from the two camera views into a 1024-dimensional latent vector $\mathbf{z}_t^{\text{cat}}$. We compute its local temporal mean using a window of $|\mathcal{W}_t| = 8$, matching the window used during TC-LeWM training.

Latent geometry and temporal stability. Figures 3(A,D) show that Raw LeWM concentrates latent norms on a thin shell, consistent with its isotropic-Gaussian marginal constraint, whereas TC-LeWM exhibits a broader and clearly multimodal distribution of latent norms. TC-LeWM also exhibits less short-timescale variation: The residual-energy fraction, as defined in Appendix A.3, is 2% for TC-LeWM, compared with 24% for Raw LeWM and 4–5% for the demonstrated robot joint positions at the same temporal scale.

The PCA projections in (B,E) further reveal extensive task overlap under Raw LeWM, whereas TC-LeWM organizes the representations into contiguous task- and scene-related structures along a curved low-dimensional manifold. The top two global principal components explain 93% of the variance for TC-LeWM, indicating that the displayed projection preserves most of its global geometry. Together, these observations are consistent with removing the direct center-contraction pressure identified in Section 3.2, allowing meaningful low-frequency structure to emerge while reducing excessive short-timescale variation. Appendix A.3 further shows that tasks occupying nearby regions of the TC-LeWM latent space often share similar visual characteristics.

Intra-task temporal structure. Figures 3(C,F) show per-task PCA projections, colored by normalized episode progress. PCA is fitted independently for each task, and the first two components explain 90% of the variance on average for TC-LeWM, compared with 31% for Raw LeWM.

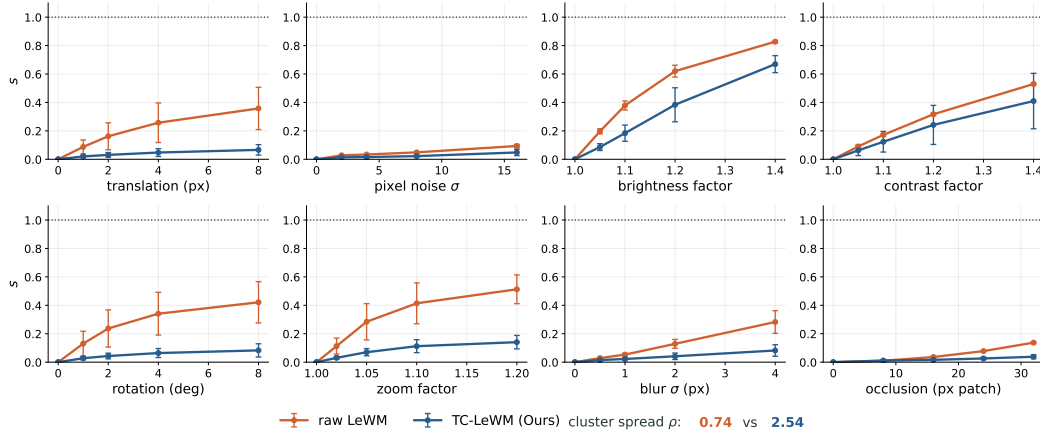


Figure 4: Relative latent displacement under eight visual perturbations on LIBERO-Long. Displacement is normalized by the median distance between distinct clean observations, such that $s = 1$ denotes the typical clean-state separation. Lines and error bars show the mean and ± 1 standard deviation over three full-pipeline training seeds. TC-LeWM exhibits smaller relative displacement across all perturbation families. Bottom annotations report the episode-level normalized cluster spread ρ .

Raw LeWM exhibits fragmented trajectories with substantial overlap between different progress stages. In contrast, TC-LeWM exhibits a more continuous and progress-aligned organization of latent states, such that nearby representations more consistently correspond to nearby stages of task execution. This more coherent temporal organization under TC-LeWM is consistent with the downstream policy receiving a more informative conditioning representation. Complete PCA projections for all ten tasks are provided in Appendix A.3.

Robustness to visual perturbations. Figure 4 evaluates the relative latent displacement induced by eight state-preserving visual perturbations. Displacement is normalized by each encoder’s median distance between distinct clean observations, such that $s = 1$ corresponds to the typical clean-state separation. Perturbation definitions and evaluated strengths are provided in Appendix A.4.

TC-LeWM exhibits smaller relative displacement across all eight perturbation families. The largest differences occur under geometric transformations, particularly translation and rotation, while substantial improvements also appear under zoom, blur, and partial occlusion. The differences are smaller for brightness and contrast, where some error bars overlap, but TC-LeWM maintains a lower mean response throughout the tested range. These results indicate improved robustness across geometric, photometric, stochastic, and occlusion-based perturbations.

The normalized cluster spread is also substantially larger for TC-LeWM than for Raw LeWM ($\rho = 2.54$ versus 0.74). This observation is consistent with perturbation-induced displacements occupying a larger fraction of the available clean-state separation in the more compressed geometry of Raw LeWM.

Such sensitivity can expose the downstream policy to out-of-distribution conditioning during closed-loop execution, allowing action errors to compound over time.

6 CONCLUSION

We identified a multi-task failure mode of marginal SIGReg and introduced TC-LeWM, which regularizes temporally centered residuals instead of the full latent marginal. Across single-task, suite-wise 10-task, and unified 40-task LIBERO settings, TC-LeWM consistently improves downstream control and learns more structured, temporally stable, and perturbation-robust representations. Future work will investigate scaling TC-LeWM to larger reward-free datasets and integrating it with Vision-Language Action Models or World-Action-Models for open-world manipulation.

ACKNOWLEDGMENTS

This work was supported in part by JST PRESTO, Japan, under Grant JPMJPR25T5.

REFERENCES

- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 15619–15629, 2023.
- Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zholus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khali-dov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025.
- Randall Balestriero and Yann LeCun. How learning by reconstruction produces uninformative features for perception. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 2566–2585. PMLR, 2024.
- Randall Balestriero and Yann LeCun. LeJEPA: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025.
- Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations*, 2022.
- Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mido Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. Featured Certification.
- Xinlei Chen and Kaiming He. Exploring simple siamese representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15750–15758, 2021.
- Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin C. M. Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea, July 2023. doi: 10.15607/RSS.2023.XIX.026.
- Anna Dawid and Yann LeCun. Introduction to latent variable energy-based models: A path toward autonomous machine intelligence. *Journal of Statistical Mechanics: Theory and Experiment*, 2024(10):104011, 2024. doi: 10.1088/1742-5468/ad292b.
- Fei Deng, Ingook Jang, and Sungjin Ahn. DreamerPro: Reconstruction-free model-based reinforcement learning with prototypical representations. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 4956–4975. PMLR, 2022.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.
- T. W. Epps and Lawrence B. Pulley. A test for normality based on the empirical characteristic function. *Biometrika*, 70(3):723–726, 1983. doi: 10.1093/biomet/70.3.723.

-
- Xiang Fu, Ge Yang, Pulkit Agrawal, and Tommi Jaakkola. Learning task informed abstractions. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 3480–3491. PMLR, 2021.
- Dibya Ghosh, Homer Rich Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, Jianlan Luo, You Liang Tan, Lawrence Yunliang Chen, Quan Vuong, Ted Xiao, Pannag R. Sanketi, Dorsa Sadigh, Chelsea Finn, and Sergey Levine. Octo: An open-source generalist robot policy. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024. doi: 10.15607/RSS.2024.XX.090.
- Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Rémi Munos, and Michal Valko. Bootstrap your own latent - a new approach to self-supervised learning. In *Advances in Neural Information Processing Systems*, volume 33, pp. 21271–21284, 2020.
- David Ha and Jürgen Schmidhuber. Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems*, volume 31, pp. 2455–2467, 2018.
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pp. 2555–2565. PMLR, 2019.
- Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations*, 2020.
- Danijar Hafner, Timothy P. Lillicrap, Mohammad Norouzi, and Jimmy Ba. Mastering atari with discrete world models. In *International Conference on Learning Representations*, 2021.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640:647–653, 2025. doi: 10.1038/s41586-025-08744-2.
- Miles Hutson, Isaac Kauvar, and Nick Haber. Policy-shaped prediction: Avoiding distractions in model-based reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 37, pp. 13124–13148, 2024. doi: 10.52202/079017-0418.
- Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan P. Foster, Pannag R. Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. Open-VLA: An open-source vision-language-action model. In *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pp. 2679–2713. PMLR, 2025.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=PqvMRDCJT9t>.
- Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. Libero: Benchmarking knowledge transfer for lifelong robot learning. *Advances in Neural Information Processing Systems*, 36:44776–44791, 2023a.
- Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with Rectified Flow. In *International Conference on Learning Representations*, 2023b. URL <https://openreview.net/forum?id=XVjTT1nw5z>.
- Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorld-Model: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026.
- William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 4195–4205, 2023.

-
- Julian Schrittwieser, Ioannis Antonoglou, Thomas Hubert, Karen Simonyan, Laurent Sifre, Simon Schmitt, Arthur Guez, Edward Lockhart, Demis Hassabis, Thore Graepel, Timothy Lillicrap, and David Silver. Mastering atari, go, chess and shogi by planning with a learned model. *Nature*, 588 (7839):604–609, 2020. doi: 10.1038/s41586-020-03051-4.
- Uladzislau Sobal, Wancong Zhang, Kyunghyun Cho, Randall Balestriero, Tim G. J. Rudner, and Yann LeCun. Learning from reward-free offline data: A case for planning with latent dynamics models. In *Advances in Neural Information Processing Systems*, volume 38, 2025.
- Shenghua Wan, Yucen Wang, Minghao Shao, Ruying Chen, and De-Chuan Zhan. SeMAIL: Eliminating distractors in visual imitation via separated models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 35426–35443. PMLR, 2023.
- Yucen Wang, Shenghua Wan, Le Gan, Shuai Feng, and De-Chuan Zhan. AD3: Implicit action is the key for world models to distinguish the diverse visual distractors. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 51546–51568. PMLR, 2024.
- Manuel Watter, Jost Springenberg, Joschka Boedecker, and Martin Riedmiller. Embed to control: A locally linear latent dynamics model for control from raw images. In *Advances in Neural Information Processing Systems*, volume 28, pp. 2746–2754, 2015.
- Philipp Wu, Alejandro Escontrela, Danijar Hafner, Pieter Abbeel, and Ken Goldberg. Daydreamer: World models for physical robot learning. In *Proceedings of The 6th Conference on Robot Learning*, volume 205 of *Proceedings of Machine Learning Research*, pp. 2226–2240. PMLR, 2023.
- Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 12310–12320. PMLR, 2021.
- Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained visual features enable zero-shot planning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 79115–79135. PMLR, 2025.
- Chuning Zhu, Max Simchowitz, Siri Gadipudi, and Abhishek Gupta. RePo: Resilient model-based reinforcement learning by regularizing posterior predictability. In *Advances in Neural Information Processing Systems*, volume 36, pp. 32445–32467, 2023.

A APPENDIX

A.1 MODEL AND IMPLEMENTATION DETAILS

World-model objective. For each camera view $k \in \{1, \dots, K_{\text{cam}}\}$, the parameter-shared encoder independently produces

$$\mathbf{z}_t^{(k)} = f_\theta(\mathbf{x}_t^{(k)}). \quad (7)$$

We denote the collection of per-view latent tokens at time t by

$$\mathbf{Z}_t \triangleq [\mathbf{z}_t^{(1)}, \dots, \mathbf{z}_t^{(K_{\text{cam}})}]. \quad (8)$$

Following the notation of LeWM, the latent prediction objective is written as

$$\hat{\mathbf{z}}_{t+1} = g_\psi(\mathbf{Z}_t, \mathbf{a}_t), \quad \mathcal{L}_{\text{pred}} \triangleq \frac{1}{K_{\text{cam}}} \sum_{k=1}^{K_{\text{cam}}} \left\| \hat{\mathbf{z}}_{t+1}^{(k)} - \mathbf{z}_{t+1}^{(k)} \right\|_2^2. \quad (9)$$

The predictor jointly processes all camera tokens, such that each predicted per-view latent may depend on observations from all camera views.

SIGReg. Given a batch of representations $\mathbf{Y} \in \mathbb{R}^{B \times D}$, SIGReg samples J random unit directions $\mathbf{u}^{(j)} \in \mathbb{S}^{D-1}$ and computes

$$\text{SIGReg}(\mathbf{Y}) \triangleq \frac{1}{J} \sum_{j=1}^J T_{\text{EP}}(\mathbf{Y} \mathbf{u}^{(j)}) \quad (10)$$

where T_{EP} is the univariate Epps–Pulley statistic. For projected samples $\mathbf{h} = \{h_b\}_{b=1}^B$, it is defined as

$$T_{\text{EP}}(\mathbf{h}) \triangleq B \int w(s) \left| \hat{\phi}_B(s; \mathbf{h}) - e^{-s^2/2} \right|^2 ds, \quad \hat{\phi}_B(s; \mathbf{h}) = \frac{1}{B} \sum_{n=1}^B e^{ish_n}, \quad (11)$$

with Gaussian weighting function $w(s)$. We use the same numerical implementation as LeWM, with $J = 1024$ random projections.

For sequential multi-view inputs, SIGReg is evaluated independently for each temporal position and camera view:

$$\mathcal{L}_{\text{SIGReg}}(\mathcal{Y}) \triangleq \frac{1}{TK_{\text{cam}}} \sum_{t=1}^T \sum_{k=1}^{K_{\text{cam}}} \text{SIGReg}(\mathcal{Y}_t^{(k)}), \quad (12)$$

where $\mathcal{Y}_t^{(k)}$ denotes the batch of representations at temporal position t from camera view k . The two camera views are not concatenated when computing SIGReg.

Raw and temporally centered objectives. Raw LeWM applies SIGReg directly to the per-view latent marginals:

$$\mathcal{L}_{\text{Raw}} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{SIGReg}}(\mathcal{Z}). \quad (13)$$

For TC-LeWM, temporal centering is performed independently for each camera view:

$$\bar{\mathbf{z}}_t^{(k)} = \frac{1}{|\mathcal{W}_t|} \sum_{s \in \mathcal{W}_t} \mathbf{z}_s^{(k)}, \quad \mathbf{r}_t^{(k)} = \mathbf{z}_t^{(k)} - \bar{\mathbf{z}}_t^{(k)}. \quad (14)$$

TC-LeWM applies the same SIGReg loss to the residuals:

$$\mathcal{L}_{\text{TC}} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{SIGReg}}(\mathcal{R}). \quad (15)$$

Thus, Raw LeWM and TC-LeWM share the same prediction objective and multi-view treatment, and differ in the SIGReg target: $\mathcal{Z}$ versus $\mathcal{R}$.

Encoder and predictor implementation. Raw LeWM and TC-LeWM use identical encoder and predictor architectures, summarized in Table 4. The two methods differ in whether SIGReg is applied to the full latent or to the temporally centered residual. For both methods, world-model checkpoints are selected using the validation inverse-dynamics-model (iDM) metric, which measures the predictability of actions from consecutive latent states.

Table 4: Encoder and latent-predictor architectures.

Module	Hyperparameter	Configuration
Encoder	Backbone	ViT-S/16 (from scratch)
	Input views	2, weight-shared, per-view tokenization
	Patch size / depth / width	16×16 / 12 / 384
	Attention heads	6
	Token projection	Linear, 384 → 512
	Embedding head	BN-MLP, 512 → 2048 → 512, in-stream
	Camera embedding	Added to [CLS]
	Frame latent	2 per-view [CLS] tokens, 512-D each
Predictor	Backbone	Block-causal Transformer, AdaLN-Zero
	Temporal context	8-frame sequence with learned temporal embeddings
	Width / depth	512 / 6
	Attention	16 heads × 64-D, inner width 1024
	Action input	Stacked actions, <code>frame_gap</code> × 7
	Action conditioning	AdaLN-Zero
	Prediction target	Next-frame per-view [CLS] tokens

Downstream behavior-cloning policy. After world-model training, the encoder is frozen and used to train a task-conditioned flow-matching policy. For each observation, the policy receives 34 visual tokens from two camera views: one [CLS] token and a 4×4 pooled patch grid per view. These tokens are used as cross-attention context and gradients are not propagated into the encoder. No proprioceptive input is used.

The policy generates an eight-step action chunk using a 12-layer DiT with AdaLN-Zero conditioning (Peebles & Xie, 2023). The noisy action tokens form the query sequence, while the frozen visual tokens provide the cross-attention keys and values. Flow time and task identity jointly condition all Transformer blocks.

We train the policy using rectified flow matching (Lipman et al., 2023; Liu et al., 2023b).

Given a demonstrated action chunk $\mathbf{a}$, Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\tau \sim \mathcal{U}(0, 1)$, the rectified flow objective is

$$\mathbf{x}_\tau = (1 - \tau)\epsilon + \tau\mathbf{a}, \quad \mathcal{L}_{\text{FM}} = \mathbb{E} \left[\|v_\phi(\mathbf{x}_\tau, \tau \mid \mathbf{H}, i) - (\mathbf{a} - \epsilon)\|_2^2 \right], \quad (16)$$

where $\mathbf{H}$ denotes the frozen visual tokens and i the task identity. At inference, the complete action chunk is generated from Gaussian noise using ten Euler steps. The eight actions are executed open loop before replanning from a new observation. Raw LeWM and TC-LeWM use identical policy architectures and training protocols. More details are provided in Table 5.

A.2 MONTE CARLO ANALYSIS DETAILS

Projected mixture model. Following the projected decomposition introduced in the main text,

$$x = \mathbf{w}^\top \mathbf{z}_t = \underbrace{\mathbf{w}^\top \bar{\mathbf{z}}_t}_c + \underbrace{\mathbf{w}^\top \mathbf{r}_t}_\epsilon, \quad (17)$$

we model the low-frequency projected component as a balanced, homoscedastic K -component Gaussian mixture:

$$p(c) = \frac{1}{K} \sum_{k=1}^K \mathcal{N}(c; \mu_k, \tau^2). \quad (18)$$

The projected residual is modeled as independent Gaussian variation,

$$\epsilon \sim \mathcal{N}(0, \sigma_r^2), \quad \epsilon \perp c. \quad (19)$$

Since $x = c + \epsilon$, the resulting projected latent marginal is

$$p(x) = \frac{1}{K} \sum_{k=1}^K \mathcal{N}(x; \mu_k, \sigma^2), \quad \sigma^2 = \tau^2 + \sigma_r^2. \quad (20)$$

Table 5: Stage-2 flow-matching policy architecture. Raw LeWM and TC-LeWM use identical configurations.

Module	Hyperparameter	Configuration
Visual condition	Latent tokens	34; 2 views $\times$ ([CLS] + 4 $\times$ 4 grid)
	Token dimension	512
	Encoder	Frozen
	Proprioception	None
DiT backbone	Width / depth	512 / 12
	Attention heads	8
	Block	Self-attention $\rightarrow$ cross-attention $\rightarrow$ FFN
	FFN ratio	4
	Conditioning	Flow-time embedding + task-ID embedding
	Modulation	AdaLN-Zero
	Parameters	79.3M
Action generation	Horizon	8
	Action dimension	7
	Normalization	Per-dimension z-score
	Objective	Rectified flow matching, MSE
	Sampling	10 Euler steps
	Execution	8-step open loop, then replan



Figure 5: First frame from the first rollout of each of the ten LIBERO-Long tasks. Tasks with similar visual characteristics tend to occupy nearby regions in the TC-LeWM latent-space projection shown in Figure 3(B,E).

Here, μ_k denotes the projected center of component k , and σ is the shared within-component standard deviation. The balanced, homoscedastic construction isolates the effect of component-center geometry on the EP objective.

Randomized center construction. To construct randomized center configurations, we first sample

$$\tilde{\mu}_k^{(m)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}(-1, 1), \quad k = 1, \dots, K. \quad (21)$$

We then standardize the sampled centers across components to obtain $\{\hat{\mu}_k^{(m)}\}_{k=1}^K$ with zero mean and unit standard deviation. This removes arbitrary translation and scale while preserving the relative spacing pattern of the sampled centers.

For each normalized configuration, we construct the component centers at spread ρ as

$$\mu_k^{(m)}(\rho) \triangleq \rho \sigma \hat{\mu}_k^{(m)}, \quad \frac{\text{std}_k [\mu_k^{(m)}(\rho)]}{\sigma} = \rho. \quad (22)$$

Thus, $\{\hat{\mu}_k^{(m)}\}_{k=1}^K$ specifies the normalized geometry of configuration m , whereas ρ controls its global component-center spread relative to the common within-component standard deviation σ . By fixing σ and sweeping ρ for the same normalized configuration, we isolate the dependence of the EP objective on relative component separation.

Monte Carlo protocol. For each $K \in \{2, 5, 10, 20\}$, we generate $M = 500$ randomized center configurations and evaluate $\rho \in \{0, 0.1, \dots, 4.0\}$. For every (K, m, ρ) , we draw $R = 20$ batches of $n = 128$ samples from the mixture in Eq. 20.

Before evaluating EP, each batch is standardized to remove its empirical location and scale:

$$\tilde{x}_i = \frac{x_i - \bar{x}}{s_x}, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s_x^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2. \quad (23)$$

The configuration-level EP objective is then computed by averaging over the R batches:

$$\mathcal{L}_{\text{EP}}^{(K,m)}(\rho) = \frac{1}{R} \sum_{r=1}^R T_{\text{EP}} \left(\tilde{\mathbf{x}}^{(K,m,r)}(\rho) \right). \quad (24)$$

This per-batch standardization yields the shape-only diagnostic reported in the main text.

Excess loss. In Figure 2, panel (a) reports the configuration-level excess loss

$$\Delta \mathcal{L}_{\text{EP}}^{(K,m)}(\rho) = \mathcal{L}_{\text{EP}}^{(K,m)}(\rho) - \mathcal{L}_{\text{EP}}^{(K,m)}(0). \quad (25)$$

The plotted center line is the median over m , and the shaded regions show the 25–75% and 5–95% percentiles.

Radial pressure. For a fixed normalized center geometry $\hat{\mu}$, let

$$\mu(\rho) = \rho \sigma \hat{\mu}, \quad \mathcal{L}_{\text{EP}}(\rho) = \mathcal{L}_{\text{EP}}(\mu(\rho)). \quad (26)$$

We define the radial pressure as

$$P(\rho) \triangleq \rho \frac{\partial \mathcal{L}_{\text{EP}}(\rho)}{\partial \rho}. \quad (27)$$

Because $\mu(\rho) = \rho \sigma \hat{\mu}$, we have $\rho \partial \mu / \partial \rho = \mu$. Therefore, by the chain rule and the linearity of the inner product,

$$\begin{aligned} P(\rho) &= \rho \frac{\partial \mathcal{L}_{\text{EP}}}{\partial \rho} \\ &= \rho \left\langle \nabla_{\mu} \mathcal{L}_{\text{EP}}, \frac{\partial \mu}{\partial \rho} \right\rangle \\ &= \left\langle \nabla_{\mu} \mathcal{L}_{\text{EP}}, \rho \frac{\partial \mu}{\partial \rho} \right\rangle = \langle \nabla_{\mu} \mathcal{L}_{\text{EP}}, \mu \rangle. \end{aligned} \quad (28)$$

Under gradient flow $d\mu/d\gamma = -\nabla_{\mu} \mathcal{L}_{\text{EP}}$,

$$\frac{d}{d\gamma} \frac{1}{2} \|\mu\|_2^2 = \left\langle \mu, \frac{d\mu}{d\gamma} \right\rangle = -\langle \mu, \nabla_{\mu} \mathcal{L}_{\text{EP}} \rangle = -P(\rho). \quad (29)$$

Therefore, $P(\rho) > 0$ indicates that the EP gradient has a contractive radial component, reducing the global component-center spread.

For a mixture with K components and randomized configuration m , we denote the corresponding objective and pressure by

$$\mathcal{L}_{\text{EP}}^{(K,m)}(\rho), \quad P_K^{(m)}(\rho) = \rho \frac{\partial \mathcal{L}_{\text{EP}}^{(K,m)}(\rho)}{\partial \rho}. \quad (30)$$

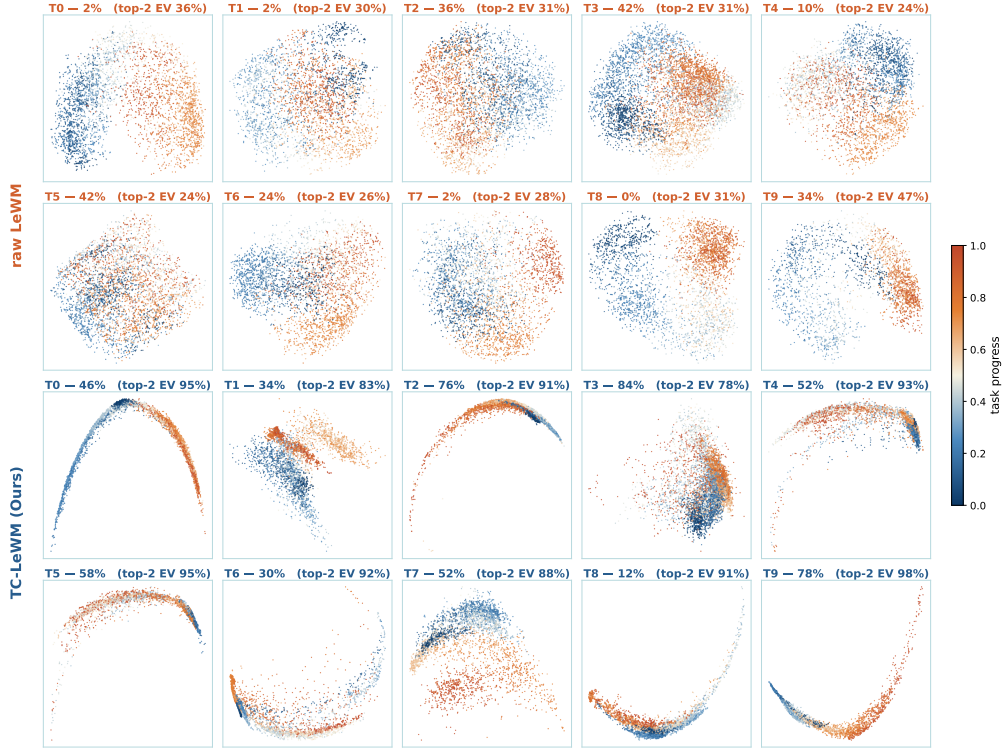


Figure 6: PCA projections of latent representations for all ten LIBERO-Long tasks from Raw LeWM and TC-LeWM models trained with the same random seed. For each task and method, PCA is fitted independently to the corresponding latent representations, and the points are colored by normalized episode progress. The explained variance of the two-dimensional projection is reported in each subplot title.

On the discrete grid $\rho_j \in \{0, 0.1, \dots, 4.0\}$, we estimate the pressure at interior points using the central difference

$$P_K^{(m)}(\rho_j) \approx \rho_j \frac{\mathcal{L}_{\text{EP}}^{(K,m)}(\rho_{j+1}) - \mathcal{L}_{\text{EP}}^{(K,m)}(\rho_{j-1})}{2\Delta\rho}, \quad \Delta\rho = 0.1. \quad (31)$$

Panel (b) reports the median of $P_K^{(m)}(\rho)$ over randomized configurations m , with the shaded regions indicating the corresponding percentile ranges.

Detectable-contraction probability. Panel (c) reports

$$q_K(\rho) = \frac{1}{M} \sum_{m=1}^M \mathbf{1} \left[P_K^{(m)}(\rho) > \delta_K \right], \quad (32)$$

where δ_K is set to the 95th percentile of the absolute radial pressure measured in the weak-signal region $\rho \leq 0.3$. Thus, $q_K(\rho)$ is the fraction of randomized configurations whose positive radial pressure exceeds the background fluctuation level.

Residual-target reference. To construct the residual-target reference curve, we divide the samples into non-overlapping windows of size $W = 8$. All samples within a window are drawn from the same mixture component. Thus, for a window $\mathcal{W}$ associated with component k , each sample can be written as

$$x_i = \mu_k(\rho) + \sigma \xi_i, \quad \xi_i \sim \mathcal{N}(0, 1), \quad i \in \mathcal{W}. \quad (33)$$

We compute the temporally centered residual as

$$r_i = x_i - \frac{1}{W} \sum_{j \in \mathcal{W}} x_j. \quad (34)$$

Substituting Eq. 33 gives

$$\begin{aligned} r_i &= \mu_k(\rho) + \sigma \xi_i - \frac{1}{W} \sum_{j \in \mathcal{W}} (\mu_k(\rho) + \sigma \xi_j) \\ &= \sigma \left(\xi_i - \frac{1}{W} \sum_{j \in \mathcal{W}} \xi_j \right). \end{aligned} \quad (35)$$

The component center $\mu_k(\rho)$ therefore cancels exactly. Consequently,

$$r_i \sim \mathcal{N} \left(0, \sigma^2 \left(1 - \frac{1}{W} \right) \right), \quad (36)$$

independently of the component-center spread ρ . Hence, varying ρ changes the full latent marginal but not the centered-residual distribution under this controlled construction. The residual-target EP curve should therefore remain constant with respect to ρ , up to finite-sample Monte Carlo variation.

A.3 ADDITIONAL LATENT REPRESENTATION ANALYSIS

Residual-energy fraction. For this analysis, we concatenate the CLS features from the two camera views and divide each episode into non-overlapping windows of $W = 8$ frames. For each latent $\mathbf{z}_t^{\text{cat}}$, we subtract the mean of its corresponding window to obtain $\mathbf{r}_t^{\text{cat}} = \mathbf{z}_t^{\text{cat}} - \bar{\mathbf{z}}_{\mathcal{W}(t)}^{\text{cat}}$.

The residual-energy fraction is defined as

$$\eta_{\text{res}} \triangleq \frac{\mathbb{E} [\|\mathbf{r}_t^{\text{cat}}\|_2^2]}{\mathbb{E} [\|\mathbf{z}_t^{\text{cat}}\|_2^2]} \quad (37)$$

where the expectation is taken over all analyzed latent samples. This quantity measures the fraction of latent energy attributable to within-window temporal variation. Using $W = 8$, we obtain 24% for Raw LeWM and 2% for TC-LeWM, compared with 4–5% for the demonstrated robot joint positions at the same temporal scale. Raw LeWM therefore exhibits roughly five to six times the within-window variation of the demonstrated joint-position reference, whereas TC-LeWM exhibits substantially lower variation, consistent with greater temporal stability.

Global task and scene organization. Panels (B) and (E) of Figure 3 show global PCA projections of the latent representations for the ten LIBERO-Long tasks. In TC-LeWM, several tasks occupy nearby regions of the latent space. For example, tasks from T1 to T3 are clustered in the top-left region, while tasks from T4 to T8 occupy the top-right region. T9 is separated from these groups in the lower-right region. Figure 5 presents the first frame from the first rollout of each task. Comparing these scenes with the PCA projection suggests that tasks occupying nearby regions often share similar visual characteristics.

Per-task PCA for all ten LIBERO-Long tasks. For each LIBERO-Long task and method, PCA is fitted independently to the corresponding latent representations. We compare Raw LeWM and TC-LeWM models trained with the same random seed, and color the projected points according to normalized episode progress.

Across the ten tasks, TC-LeWM generally exhibits clearer progress-related structure, with latent representations showing a more coherent organization along the task trajectory. In contrast, Raw LeWM often shows stronger overlap between representations from different stages of the same task. The main text presents four representative tasks for readability, while Figure 6 provides the complete comparison for all ten tasks.

Normalized episode-cluster spread. To relate the learned latent geometry to the normalized spread ρ used in the Monte Carlo analysis, we treat each LIBERO-Long training episode as an empirical context cluster. For each trained encoder, we use all 500 training episodes, retain every fourth frame, and concatenate the CLS tokens from the two camera views into a 1024-dimensional representation.

We compute ρ as the ratio between the RMS spread of the episode-level latent centers and the RMS variation of individual latents around their corresponding episode center. Episode centers are weighted by the number of retained frames. This dimensionless ratio measures context-level

Table 6: Visual perturbations used in the LIBERO-Long robustness evaluation. Listed strengths exclude the clean baseline.

Perturbation	Strengths	Application
Translation	1, 2, 4, 8 px	Horizontal shift with edge padding
Pixel noise	$\sigma = 2, 4, 8, 16$	Additive Gaussian noise on uint8 RGB values
Brightness	$\times 1.05, 1.1, 1.2, 1.4$	Global intensity scaling and clipping
Contrast	$\times 1.05, 1.1, 1.2, 1.4$	Scaling around the image mean
Rotation	1, 2, 4, 8°	In-plane rotation with edge padding
Zoom	$\times 1.02, 1.05, 1.1, 1.2$	Centered zoom and crop to 224×224
Gaussian blur	$\sigma = 0.5, 1, 2, 4$ px	Spatial Gaussian filtering
Occlusion	8, 16, 24, 32 px	Random black square patch

Table 7: Ablation of the temporal window size on LIBERO-Long. Results are closed-loop success rates (%), reported as mean $\pm$ standard deviation over three seeds. Raw LeWM applies SIGReg directly to the full latent.

Setting	$W = 4$	$W = 8$	$W = 32$	Episode	Raw LeWM
Temporal scale	0.7 s	1.4 s	~ 6 s	~ 36 s	–
Success rate	44.7 ± 4.8	51.8 ± 9.2	48.2 ± 9.1	53.4 ± 3.2	29.9 ± 8.3

separation relative to within-episode variation and is invariant to global rescaling of the latent representation.

We compute ρ independently for each of the three training seeds and then average the resulting values. Raw LeWM yields $\rho = 0.74$, whereas TC-LeWM yields $\rho = 2.54$, indicating substantially greater episode-level separation relative to within-episode variation under TC-LeWM.

A.4 VISUAL PERTURBATION DETAILS

We evaluate the robustness of the learned representations on LIBERO-Long using eight state-preserving visual perturbations: translation, pixel noise, brightness, contrast, rotation, zoom, Gaussian blur, and partial occlusion. The perturbations are applied only at evaluation time and do not modify the underlying robot state or task progress.

For each clean multi-view observation $\mathbf{x}_i = \{\mathbf{x}_i^{(k)}\}_{k=1}^{K_{\text{cam}}}$, let $\mathcal{T}_\alpha^{(k)}$ denote the perturbation applied to view k at strength α . We concatenate the corresponding per-view CLS features:

$$\mathbf{z}_i^{\text{cat}} = [f_\theta(\mathbf{x}_i^{(1)}); f_\theta(\mathbf{x}_i^{(2)})], \quad \mathbf{z}_i^{(\alpha), \text{cat}} = [f_\theta(\mathcal{T}_\alpha^{(1)}(\mathbf{x}_i^{(1)})); f_\theta(\mathcal{T}_\alpha^{(2)}(\mathbf{x}_i^{(2)}))]. \quad (38)$$

We normalize the perturbation-induced displacement by the median latent distance between distinct clean observations:

$$s_i(\alpha) = \frac{\|\mathbf{z}_i^{(\alpha), \text{cat}} - \mathbf{z}_i^{\text{cat}}\|_2}{\text{median}_{j \neq \ell} \|\mathbf{z}_j^{\text{cat}} - \mathbf{z}_\ell^{\text{cat}}\|_2}. \quad (39)$$

The normalization is computed independently for each trained encoder. Thus, $s = 1$ denotes a perturbation-induced displacement equal to the typical separation between two distinct clean observations.

Table 6 summarizes the perturbation definitions and evaluated strengths.

For each perturbation family and strength, we average $s_i(\alpha)$ over the evaluated observations. Figure 4 reports the mean and standard deviation over three independently trained full-pipeline seeds. Lower values indicate greater robustness to the corresponding visual perturbation.

A.5 ABLATION ON TEMPORAL WINDOW SIZE

TC-LeWM introduces the temporal window size W for estimating the local latent center. In principle, W ranges from one frame to the full episode length. When $W = 1$, each latent is identical

to its estimated center, making the centered residual identically zero. SIGReg therefore provides no effective regularization signal, resulting in representation collapse in our experiments. At the other extreme, episode-level centering estimates a single center using all frames in an episode.

We evaluate four non-degenerate center-estimation scales on LIBERO-Long: $W \in \{4, 8, 32\}$ and the full episode length. These settings span temporal scales from approximately 0.7 s to 36 s. All other world-model and policy-training settings are kept unchanged.

As shown in Table 7, all evaluated temporal-centering variants substantially outperform Raw LeWM. Within the tested range, performance shows no clear dependence on the temporal window size: neither shorter nor longer windows consistently perform better. The default setting $W = 8$ achieves 51.8% success, while episode-level centering obtains the highest mean success rate of 53.4%. These results indicate that, on LIBERO-Long, the effectiveness of TC-LeWM is not sensitive to the precise choice of W once the degenerate $W = 1$ case is excluded. We use $W = 8$ as the default setting in the main experiments.